SAMPLE

Earth's Dynamic Systems: The Greenhouse Effect

How Earth Stays Warm

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Introduction

The greenhouse effect is the reason Earth is warm. Without it, the planet's average surface temperature would sit far below the freezing point of water, ice would cover the tropics, and the world we know, oceans, forests, and us, could not exist. With it, the surface enjoys a comfortable average near fifteen degrees Celsius, a full thirty-three degrees warmer than the planet would be with the same sunlight and no blanket of greenhouse gases. The effect is old physics, measured in the laboratory before the American Civil War, named before the first airplane flew, and it is also the single most consequential idea in modern environmental science, because the same mechanism that keeps Earth habitable, strengthened by human industry, is now warming the planet at a pace the geological record rarely shows.

This book explains the effect whole: where the idea came from, in the work of Joseph Fourier, Eunice Newton Foote, John Tyndall, and Svante Arrhenius; how the blanket actually works, molecule by molecule and layer by layer; how scientists measure it, from mountain observatories to satellites; how it operates on Venus, Mars, and Titan, where it has run away, fizzled, and been fought by haze; what a runaway greenhouse is and why Earth is not immediately in danger of one but cannot escape one in the far future; and what the enhanced greenhouse, the human-strengthened version, is doing to the planet's energy books right now. The account uses no equations, keeps plain language throughout, and sorts carefully between what is settled, what is debated, and what remains unknown. The reader who finishes it will understand one of the few scientific ideas that genuinely changed how civilization sees itself.

Chapter 1: Fourier's Question

The story properly begins with a puzzle about temperature. In the eighteen-twenties, Joseph Fourier, the French mathematician who had made his name analyzing how heat spreads, asked a deceptively simple question: how hot should the Earth be? The arithmetic was straightforward. Sunlight arrives, the planet absorbs a share and reflects the rest, and the warmed surface radiates the energy back toward space as invisible infrared light. Balance the two flows and you can compute the temperature a planet of Earth's size and distance from the Sun ought to hold: well below the freezing point of water, colder than the coldest nights recorded anywhere on the actual Earth. The observed world, mild and liquid, was far warmer than this bare arithmetic allowed, and Fourier, publishing in 1824 and expanding in 1827, reasoned that the atmosphere must be doing something: it lets the sun's visible light through freely, but, he suspected, it does not pass the infrared radiation the warmed surface emits in return, so heat accumulates beneath it. The insight is the greenhouse effect in one sentence, and Fourier earned the founding seat in its history.

The same era supplied the first hints about which part of the air does the trapping. Around 1774, Horace Benedict de Saussure, a Swiss naturalist and Alpine explorer, showed that non-luminous warm objects emit radiant heat, building instruments to trap and measure it; his work gave the century a working idea of invisible heat rays. Building on Fourier's reasoning, the French physicist Claude Pouillet proposed in 1838 that the two gases already suspected of meteorological importance, water vapor and carbon dioxide, might be the atmosphere's infrared blockers, catching the surface's outgoing radiation and warming the sky. The proposal was physically sound but unproven; nobody had yet shown in a laboratory that any particular gas absorbed the radiant heat in question, and until that happened, the greenhouse remained an elegant hypothesis.

The proof arrived from an unexpected quarter, and from a person history nearly lost. In 1856, at a scientific meeting in Albany in the summer of 1856, a paper by Eunice Newton Foote was read on her behalf, describing a set of glass tubes she had filled with different gases, set in sunlight, and warmed with thermometers. Her finding, stated in her own memorable closing sentence, was that a sky filled with such gas would give our earth a high temperature; the warming effect of the sun was greater for moist air than for dry air, and greater for carbon dioxide than for ordinary air. Foote, an amateur scientist and women's-rights campaigner from New York, had demonstrated the greenhouse properties of carbon dioxide three years before anyone else published on the subject, and her paper, little noticed at the time, was only rediscovered and celebrated in the twenty-first century. Hers is the first experimental evidence in the field's history, and the fact that it came from outside the scientific establishment, delivered by Joseph Henry of the Smithsonian because women were not permitted to read their own work aloud, is part of the story a modern telling should keep.

Fourier's question had by then acquired a champion in Britain. John Tyndall, a gifted experimentalist at the Royal Institution in London, began in 1859 a program of measurements aimed directly at radiant heat, and the following

chapters take up his results, which converted the greenhouse from speculation to laboratory fact. The founding thread, though, deserves its closing observation: the entire framework, sunlight in, infrared out, a selective blanket overhead, was assembled by 1838 from pure reasoning and crude instruments, a generation before anyone measured a single gas's absorption. Physics sometimes works that way, the answer sketched early and the proof arriving late, and the greenhouse effect is one of its cleanest examples.

Fourier's 1827 paper also contains a sentence that history has learned to reread. The establishment and progress of human societies, he wrote, the action of natural forces, can over many centuries modify the average degree of heat of the globe, because the analysis of this equilibrium permits the effects of human industry to be perceived. He was speaking in general terms, of deforestation and land clearing, with no suspicion of fossil carbon, and the sentence lay dormant for a century; but the founding paper of greenhouse physics is also the founding paper of the idea that people can move the planetary thermostat, and the modern reader meets it with a strange double feeling of admiration and vertigo. The men and women of this founding chapter, working with brass tubes, glass jars, and thermometers read by eye, had within their grasp every concept the twenty-first century argues about: the selective atmosphere, the trace gases, the energy balance, and even the human hand on the scale.